

PROJECT BUTTON HR–V0 ELECTRICAL V3 PANEL–TOPOLOGY CANDIDATE

P1.15 logic plus explicit XD24/XD0 distribution and XN1/XN2/XN3 three–way nodes; no hidden splices.

01 External listed sources and DC boundaries

File: 01_external_sources.kicad_sch

02 Dual–channel E–stop and RESET eligibility

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03 Distinct ARM and watchdog–gated SR1 supply

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04 Contactor coils, mirror contacts and EDM

File: 04_contactor_edm.kicad_sch

05 Redundant actuator–power interruption

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06 Protected actuator branches and current–limiter carriers

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07 Independent watchdog power, controller and drivers

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08 Calculated dual–channel 24 V watchdog feedback

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09 Compute and control terminals

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10 U2D2, DXL star, actuator ports and bonding boundary

File: 10_actuator_interfaces.kicad_sch

11 Watchdog PCB external connectors

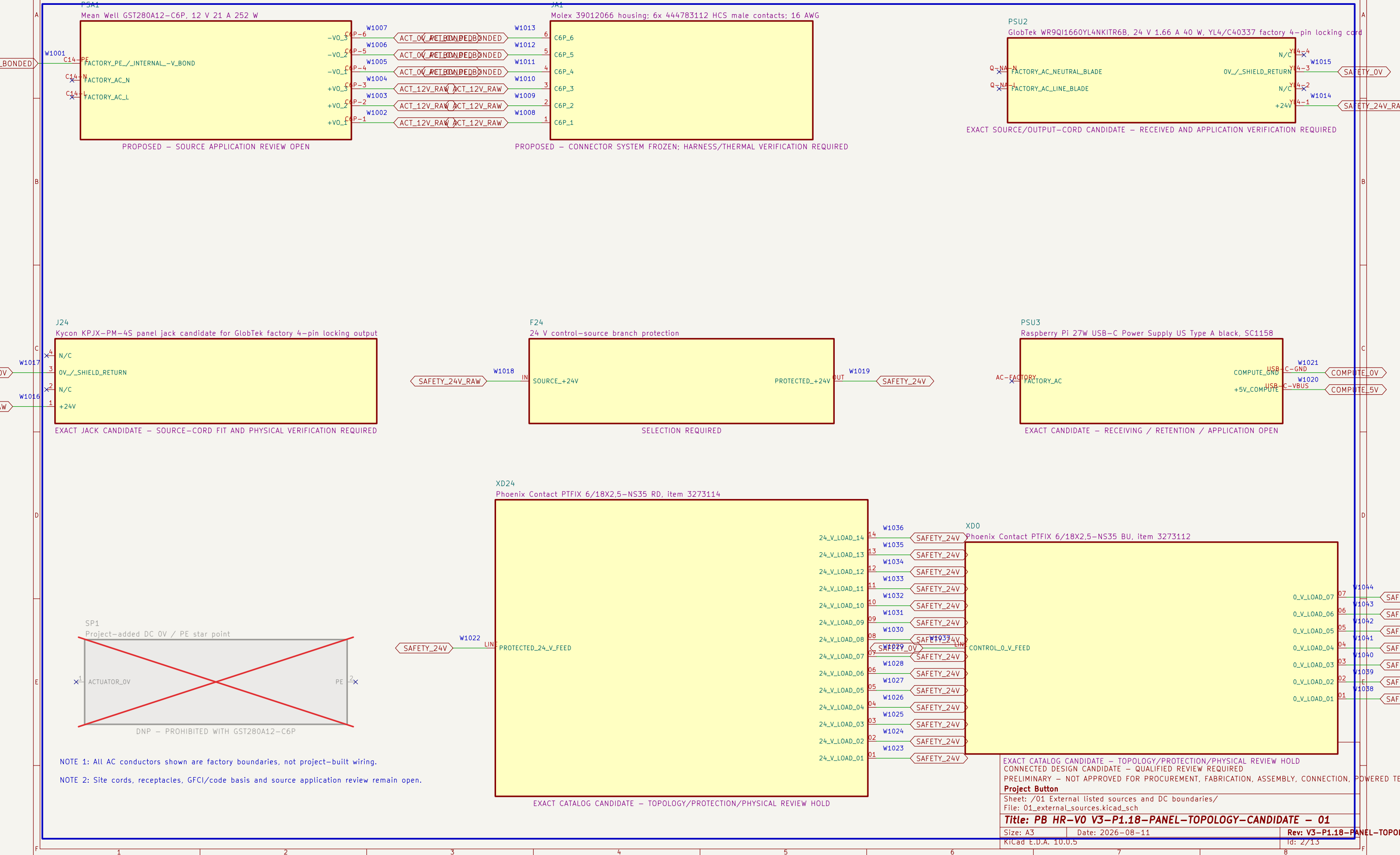
File: 11_watchdog_pcb_connectors.kicad_sch

12 Watchdog PCB test access

File: 12_watchdog_pcb_test_access.kicad_sch

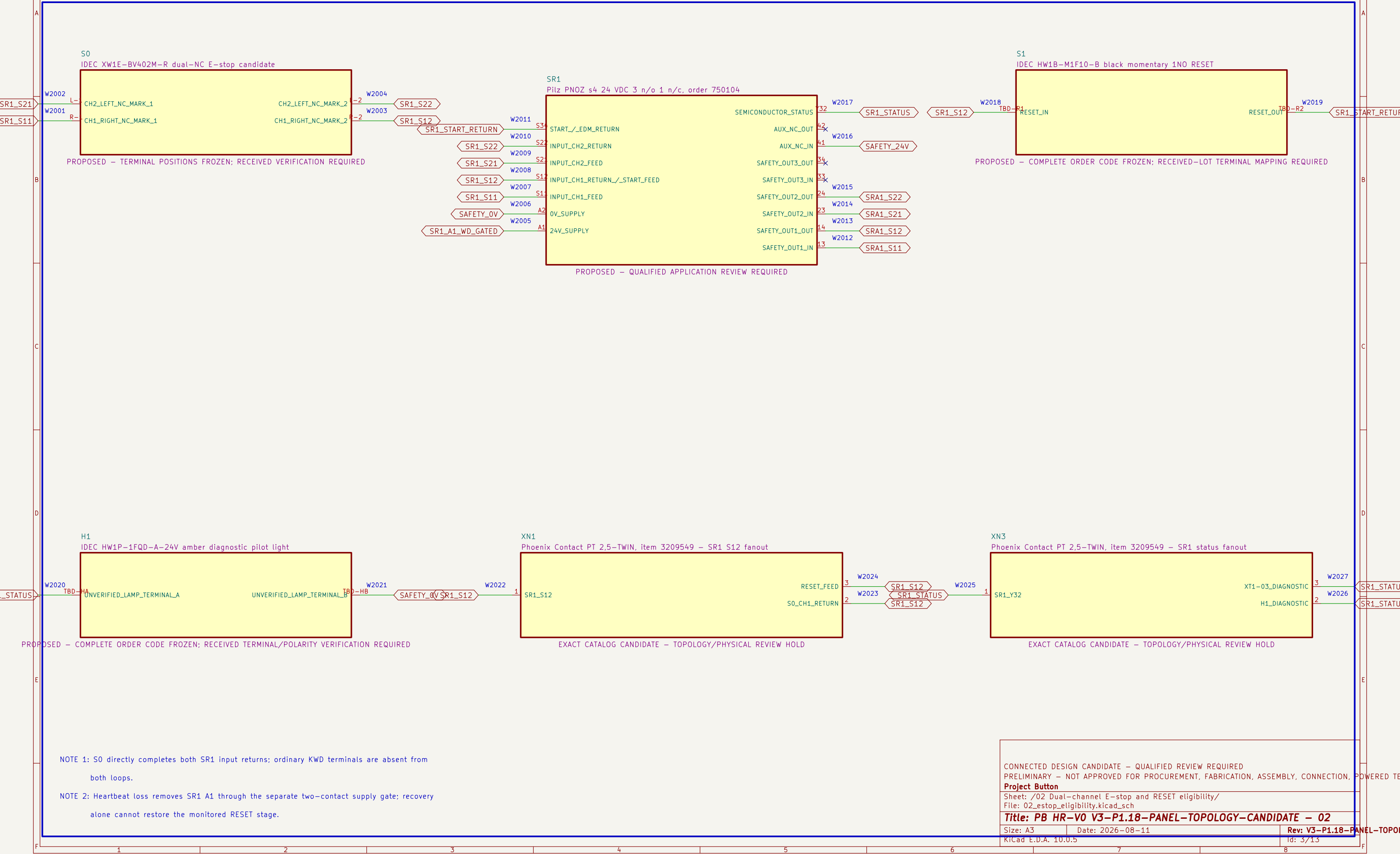
01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.



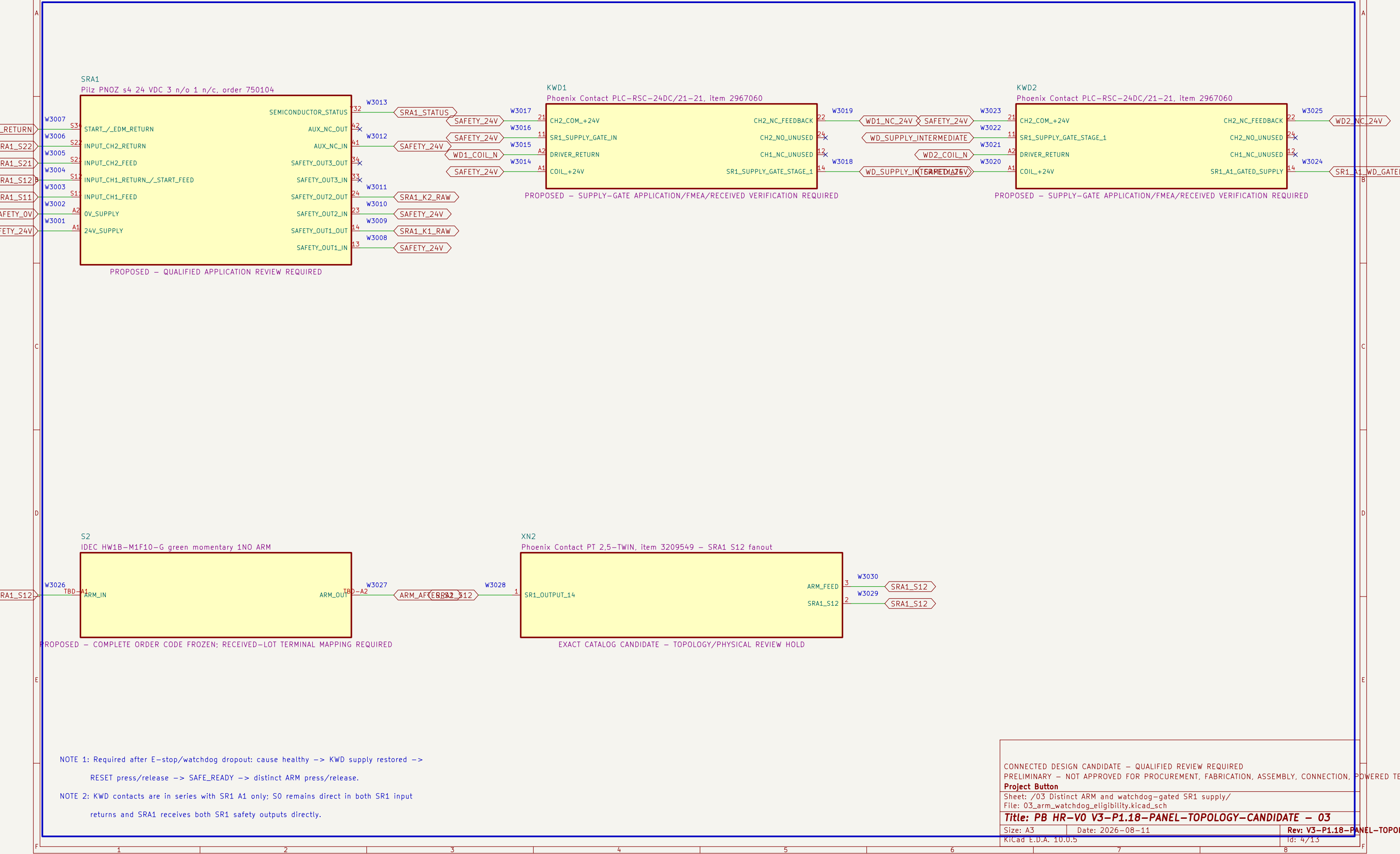
02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains only one S0 NC contact; RESET cannot energize K1/K2.



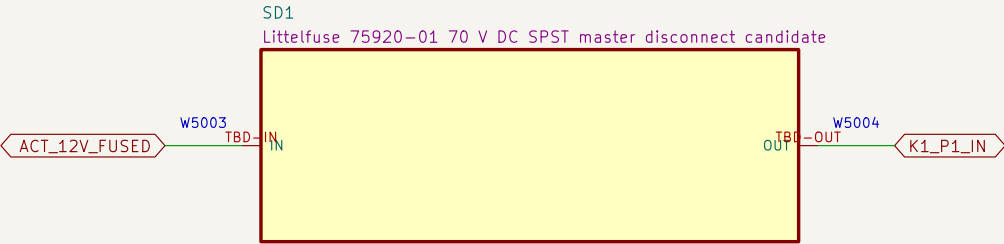
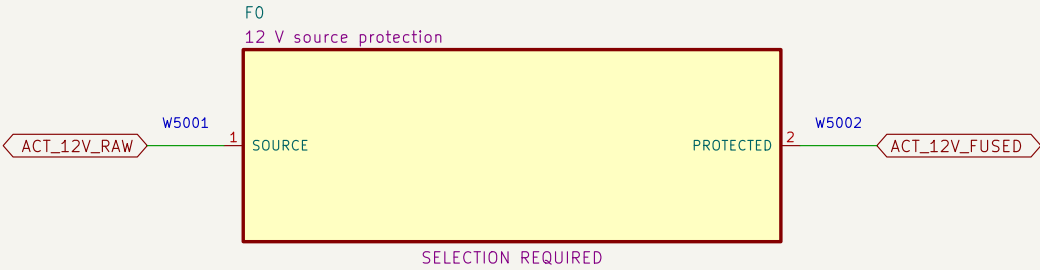
03 Distinct ARM and watchdog-gated SR1 supply

Two ordinary KWD contacts gate SR1 A1; SRA1 still requires SR1 outputs, EDM proof and a new ARM action.

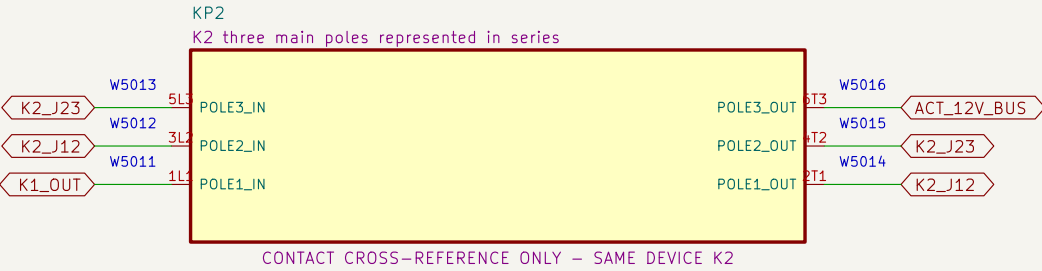
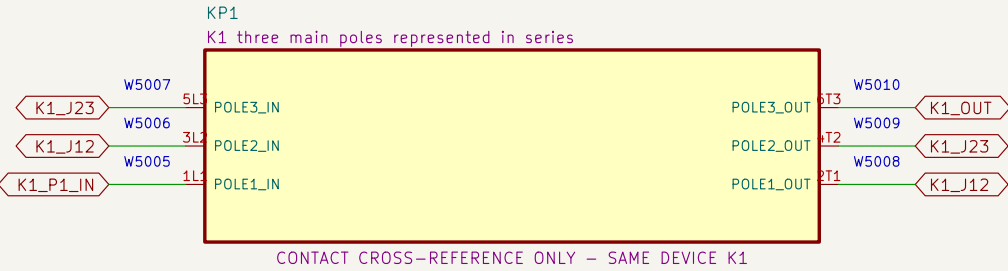


05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



PROPOSED – ORDER CODE AND SPST HIGH–SIDE TOPOLOGY FROZEN; CONDUCTOR/LUG/FAULT/LOAD–BREAK/PLACEMENT/LOCKOUT APPLICATION SELECTION REQUIRED



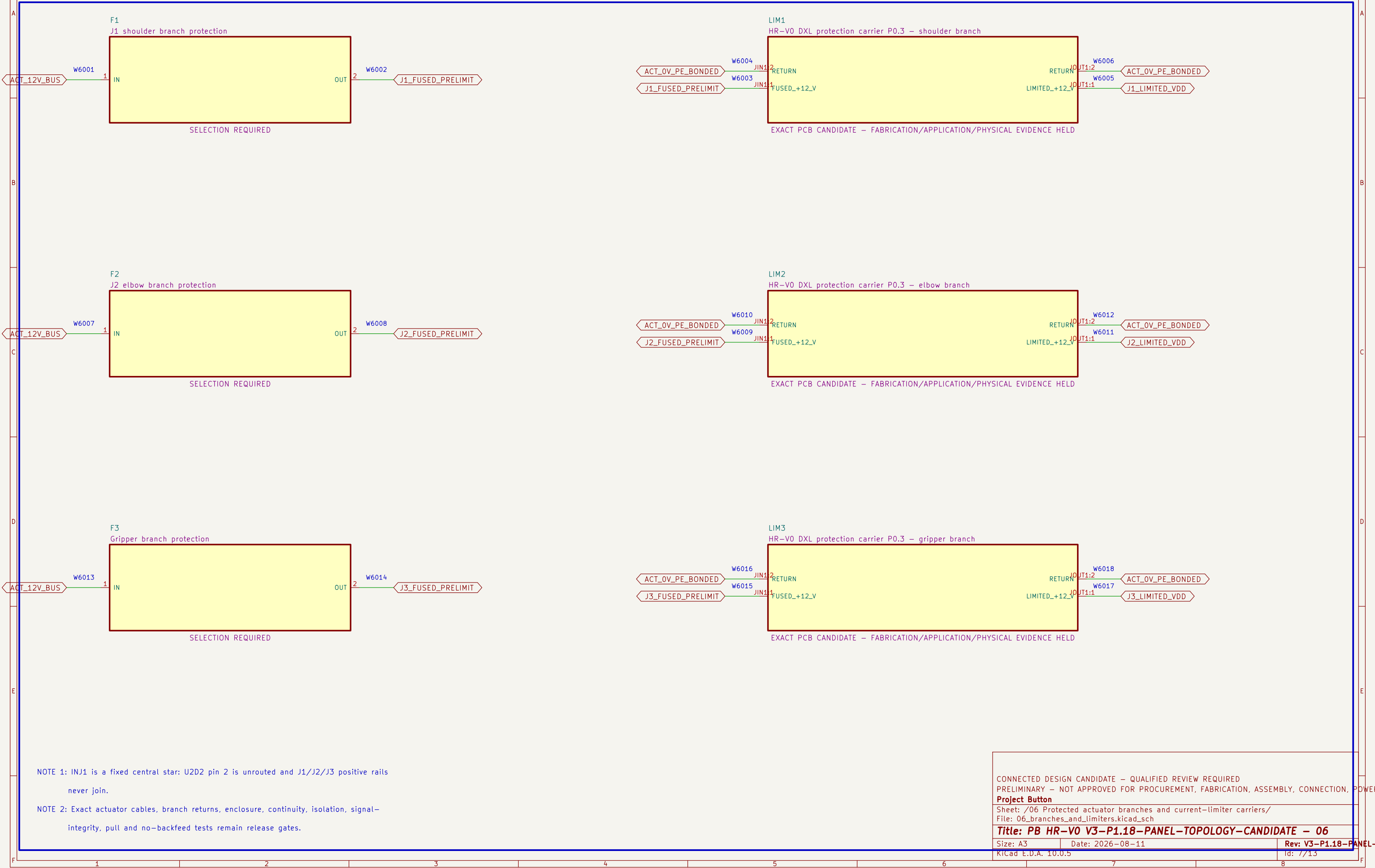
NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION		
Project Button		
Sheet: /05 Redundant actuator–power interruption/ File: 05_actuator_interruption.kicad_sch		
Title: PB HR–V0 V3–P1.18–PANEL–TOPOLOGY–CANDIDATE – 05		
Size: A3	Date: 2026–08–11	Rev: V3–P1.18–PANEL–TOPOLOGY
KiCad E.D.A. 10.0.5	Id: 6/13	

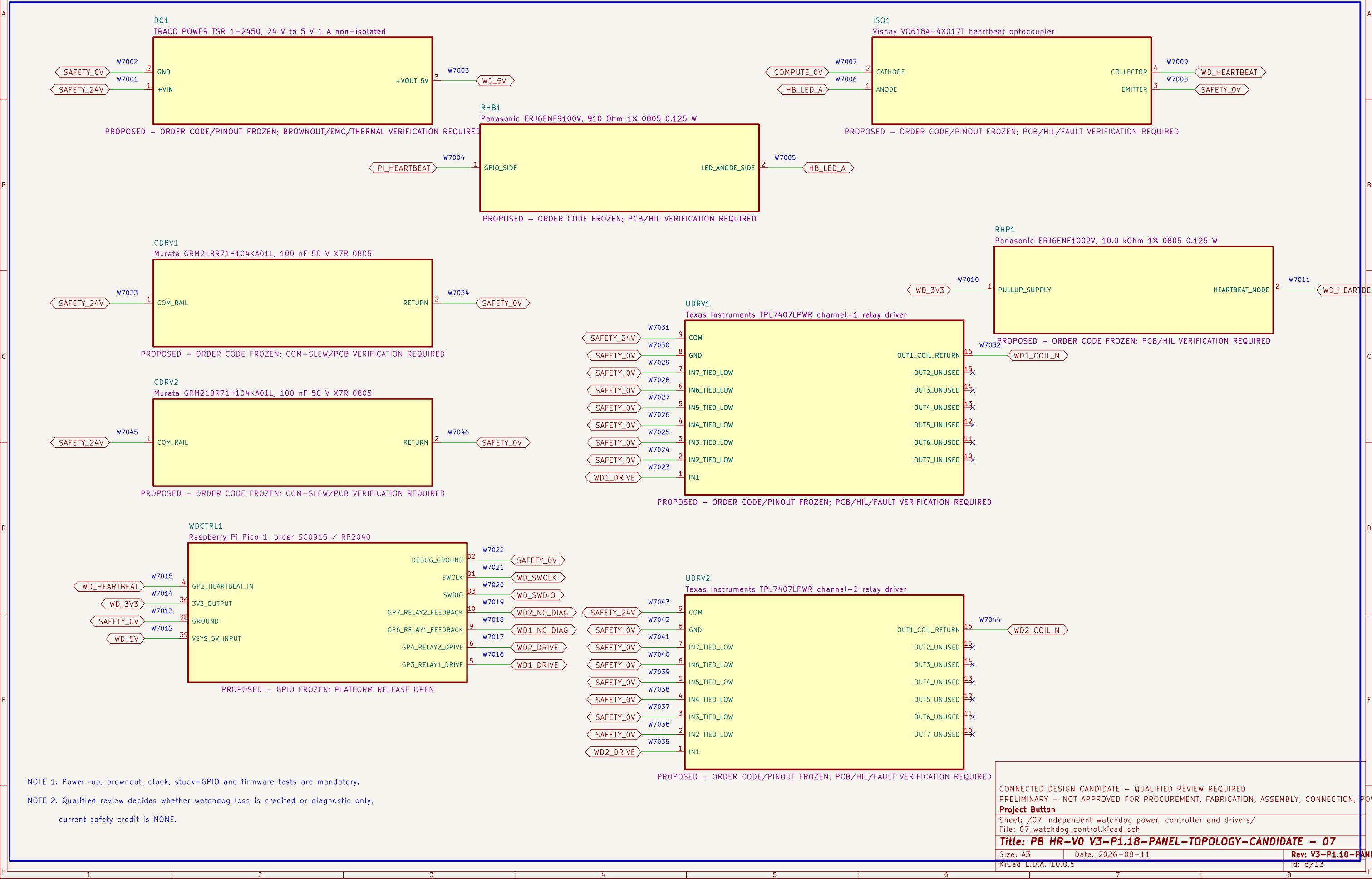
06 Protected actuator branches and current–limiter carriers

Each fuse output and carrier output has a distinct positive rail; physical protection performance remains unproved.



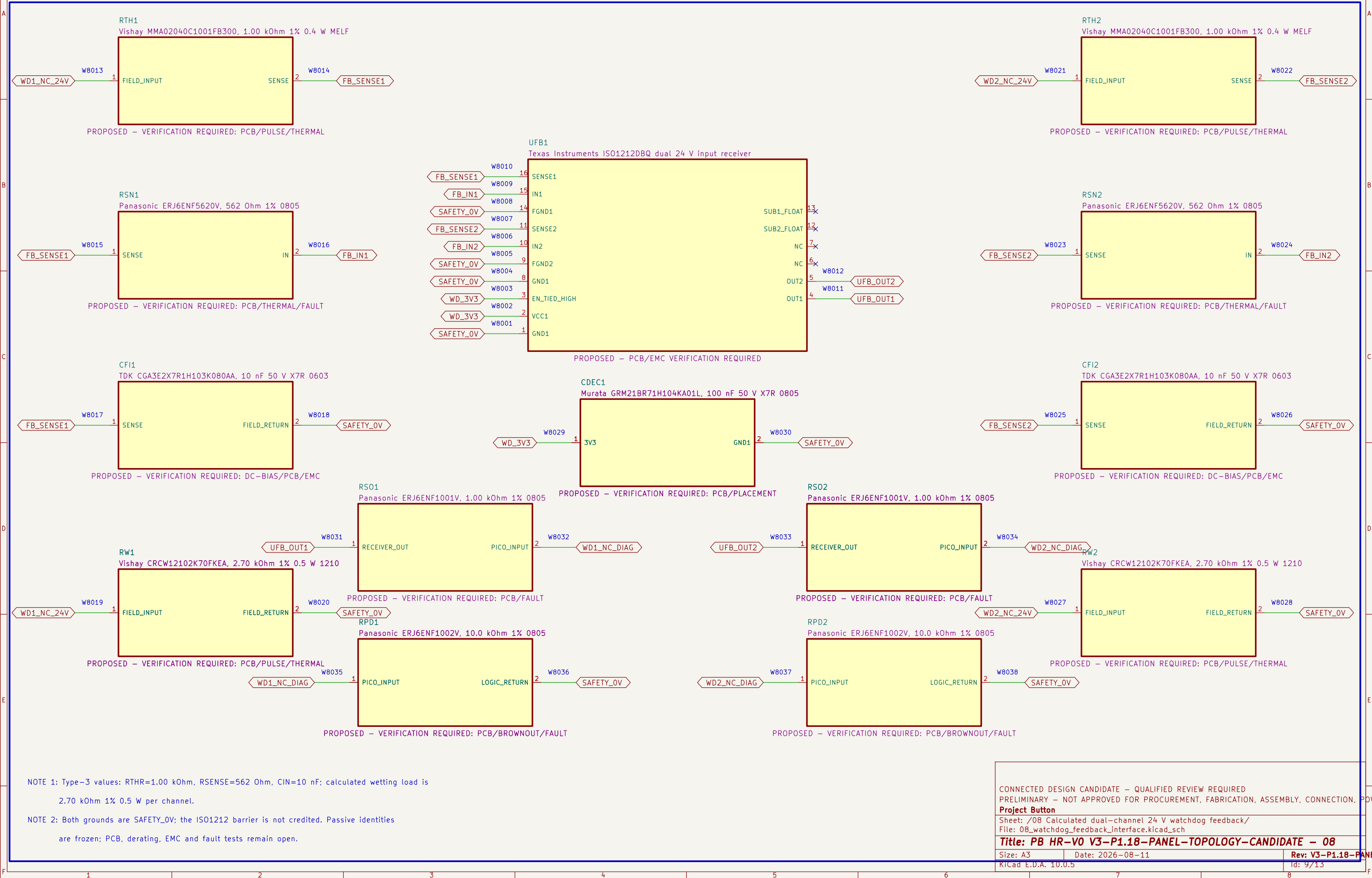
07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



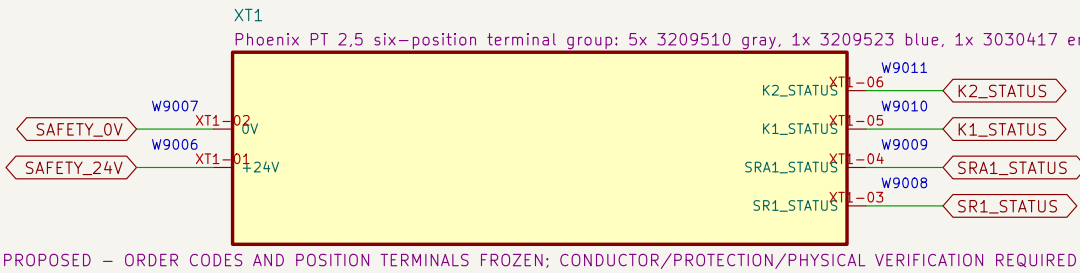
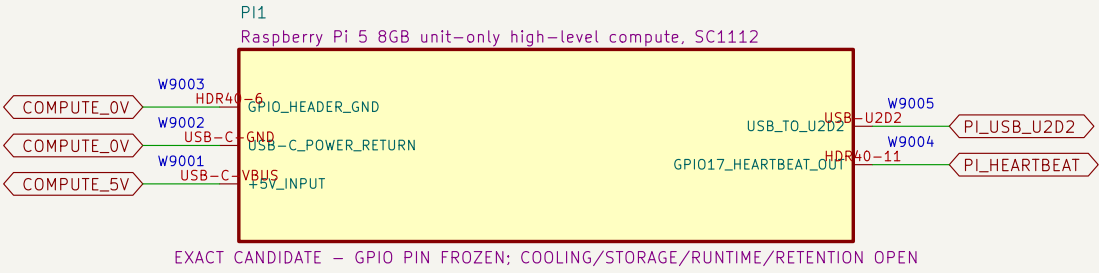
08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



09 Compute and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



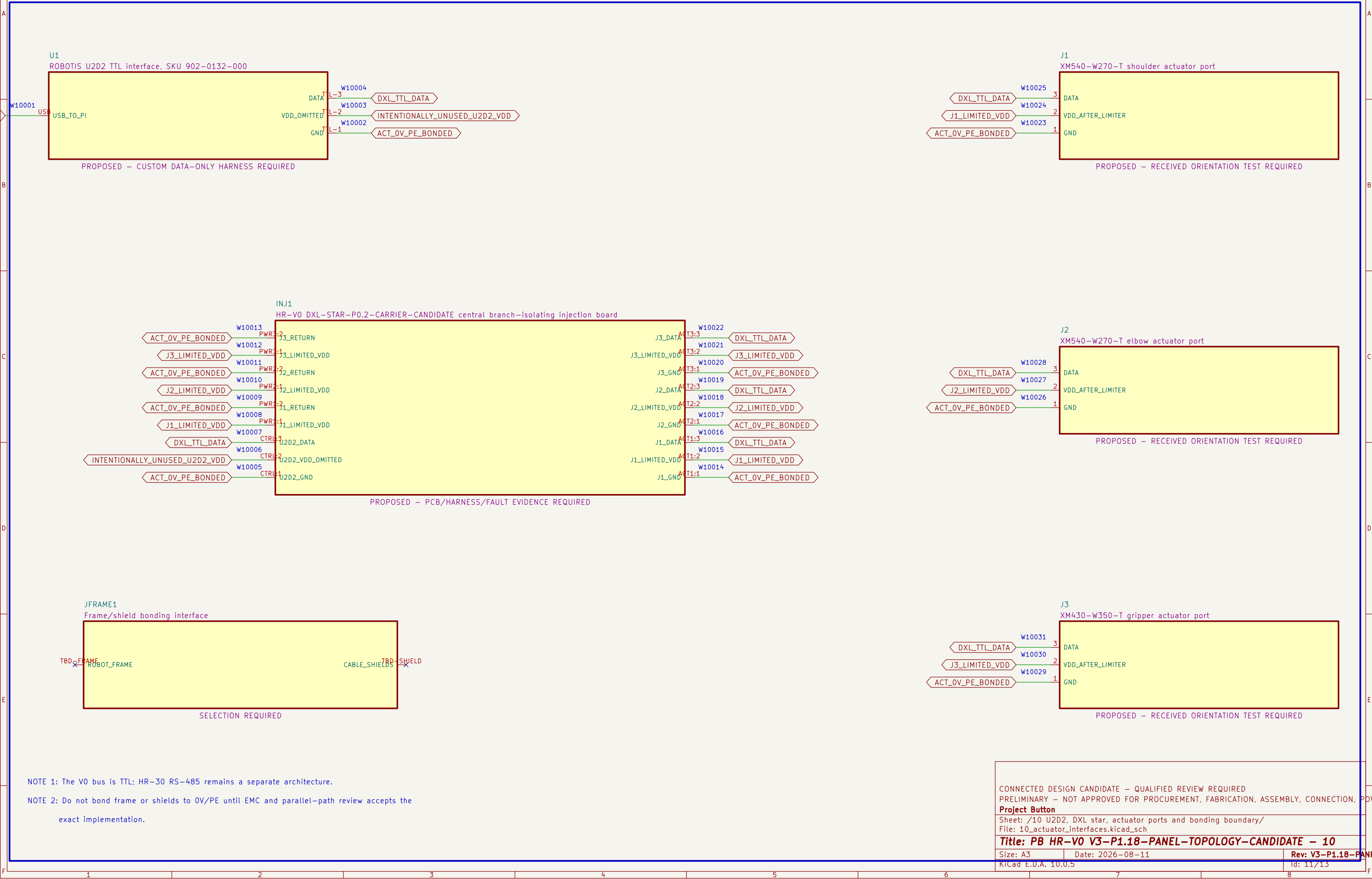
NOTE 1: No installed debug connector exists; watchdog programming uses only TP15/TP16/TP2 under a future controlled unpowered-fixture procedure.

NOTE 2: Debug connection, halt, flash and disconnect must not enable outputs, back-power the board or bypass a protective function.

NOTE 3: Heartbeat cable, GPIO runtime, timing, terminal family, markers, conductor range, torque and enclosure layout remain selection required.

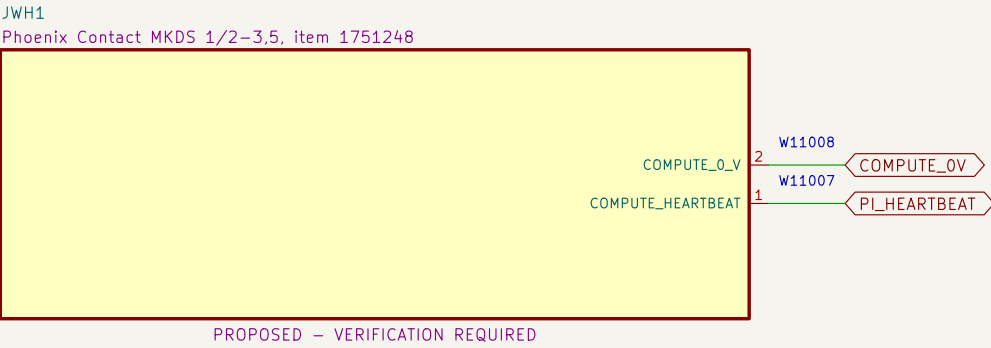
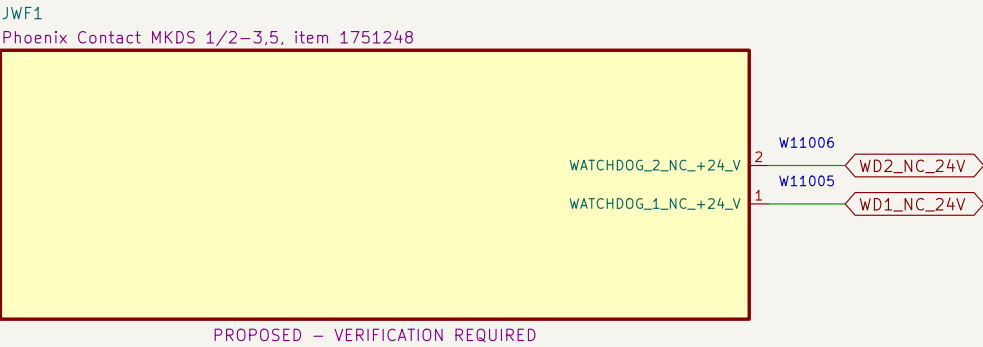
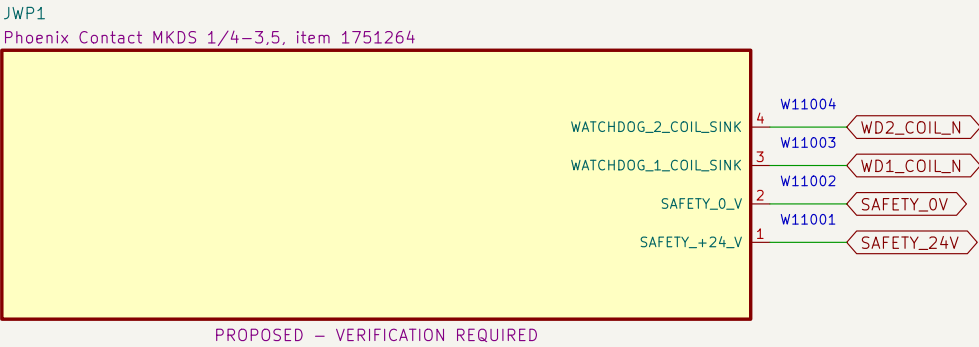
10 U2D2, DXL star, actuator ports and bonding boundary

U2D2 pin 2 is omitted; DXL–STAR accepts three distinct limited positive rails and routes DATA/return.



11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.



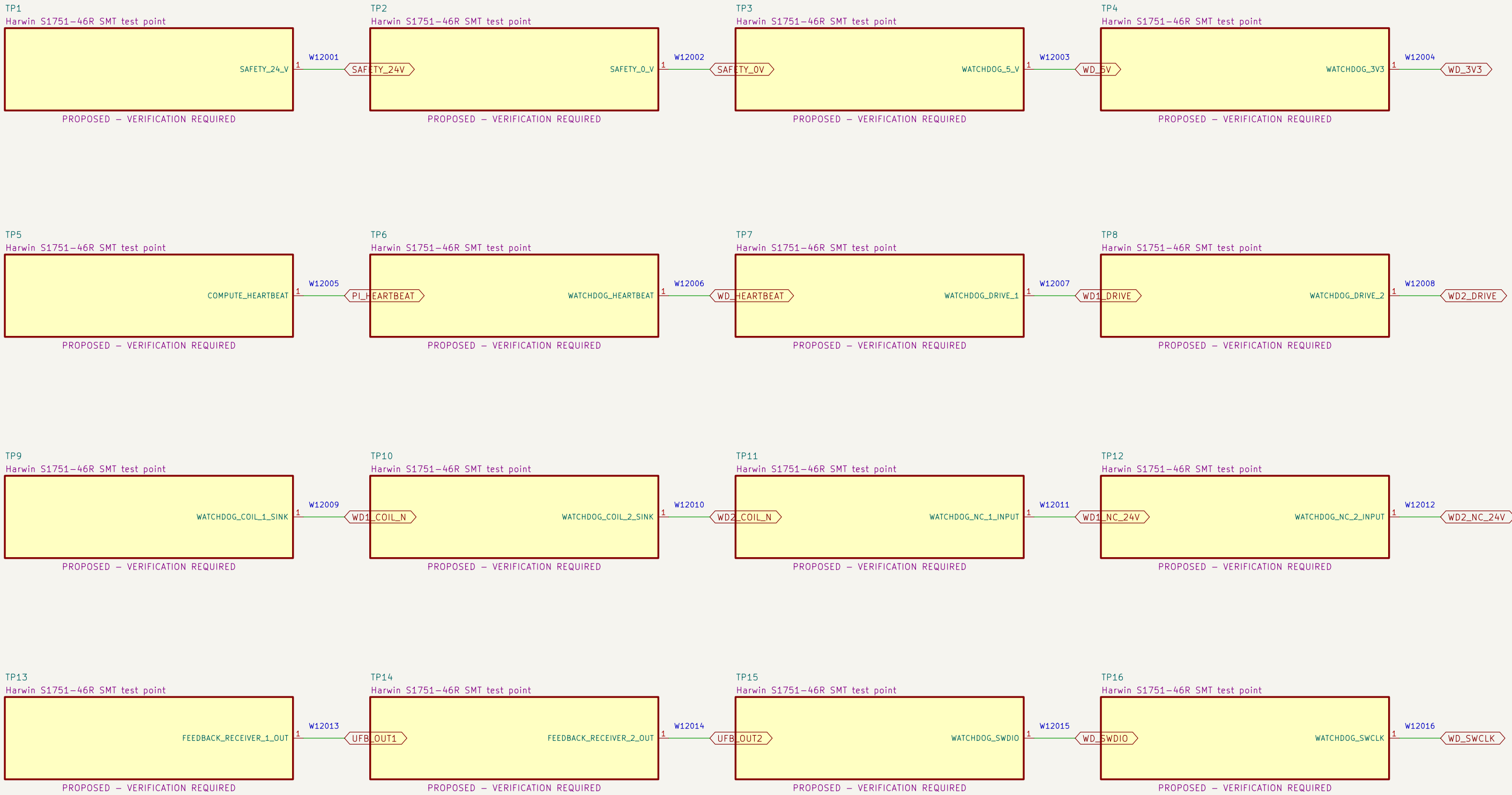
NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION		
Project Button		
Sheet: /11 Watchdog PCB external connectors/ File: 11_watchdog_pcb_connectors.kicad_sch		
Title: PB HR–V0 V3–P1.18–PANEL–TOPOLOGY–CANDIDATE – 11		
Size: A3	Date: 2026–08–11	Rev: V3–P1.18–PANEL–TOPOLOGY–CANDIDATE – 11
KiCad E.D.A. 10.0.5	Id: 12/13	

12 Watchdog PCB test access

Exact clip-compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.